

(Thm) (no proof). Let  $A_{m \times n}$  satisfy the RIP of order  $2s$  with  $\delta_{2s} < \frac{4}{\sqrt{41}}$ .

Then,  $A$  satisfies the robust null space property with certain  $\rho \in (0, 1)$  and  $\tau > 0$ , depending only on  $\delta_{2s}$ .

In particular, if  $x \in \mathbb{R}^n$  satisfies  $\|Ax - b\|_2 \leq \eta$  and if  $x^*$  is the solution of the robust basis pursuit problem then

$$\|x - x^*\|_1 \leq C_1 \sigma_s(x) + C_2 \sqrt{s} \eta$$

with certain constants  $C_1, C_2$  depending only on  $\delta_{2s}$ .



Next, return to

$$\delta_s = \max \{ \|A_s^T A_s - I\|_2 :$$

$$S \subseteq \{1, \dots, n\} \text{ with } |S| \leq s \}$$

Note that

$$(A_s^T A_s - I)_{ij} = \langle a_{k_i}, a_{k_j} \rangle - \delta_{ij}$$

$$1 \leq i, j \leq s$$

$a_j \in \mathbb{R}^m$  is the  $j$ -th column of  $A_{m \times n}$ .

- Assume from now on that all columns of  $A$  are normalized ( $\|a_j\|_2 = 1$ ). Then all diagonal elements of  $A_s^T A_s - I$  are zero.

Def Let  $A_{m \times n}$  be a matrix with normalized columns  $a_j \in \mathbb{R}^m$ ,  $1 \leq j \leq n$ .



The coherence  $\mu = \mu(A)$  of  $A$  is defined as

$$\mu := \max \{ |\langle a_i, a_j \rangle| : i, j \in \{1, \dots, n\}, i \neq j \}$$

(Thus each off-diagonal entry of  $A_s^T A_s$  has abs. value less than  $\mu$ ).

By the Gershgorin disk theorem

$$\begin{aligned} \|A_s^T A_s - I\|_2 &= |\lambda_{\max}| = |\lambda_{\max} - 0| \leq \\ &\leq \sum_{\substack{j=1 \\ j \neq i}}^s |\langle a_{k_i}, a_{k_j} \rangle| \leq (s-1)\mu \end{aligned}$$

Proposition. Let  $A_{m \times n}$ ,  $n > m$  have normalized columns and a coherence  $\mu$  satisfying  $\mu < \frac{1}{2s-1}$ . Then each  $s$ -sparse vector  $x \in \mathbb{R}^n$  is uniquely



determined by its observation

$$b := Ax.$$

Thm (Welsh bound). The coherence of a matrix  $A_{m \times n}$  with  $n > m$  satisfies

$$\mu \geq \sqrt{\frac{n-m}{m(n-1)}}$$

Pl. Consider the matrices

$$B = (A^T A)_{n \times n}$$

$$C = (A A^T)_{m \times m}$$

The entries of  $B$  are

$$B_{ij} = \langle a_i, a_j \rangle_2$$

Note also that  $\text{tr}(C) = \text{tr}(B) =$

$$= \sum_{j=1}^n \|a_j\|_2^2 = n$$



Using the Frobenius norm  
defined by the inner product

$$\langle G, H \rangle_F = \text{tr}(GH^T) = \sum_{i,j=1}^m G_{ij} H_{ij}$$

$$G, H \in \mathbb{R}^{m \times m}$$

we obtain

$$\begin{aligned} n &= \text{tr}(C) = \langle C, I \rangle_F \leq \|C\|_F \|I\|_F = \\ &= \sqrt{n} \sqrt{\text{tr}(CC^T)} \end{aligned}$$

$$\text{Next, } \text{tr}(CC^T) = \text{tr}(AA^T AA^T) =$$

$$= \text{tr}(A^T A A^T A) = \text{tr}(BB^T) = \sum_{i,j=1}^n |\langle a_i, a_j \rangle|^2$$

$$= n + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2 \leq n + n(n-1)\mu^2$$

Thus

$$n^2 \leq n(n + n(n-1)\mu^2)$$



solve for  $\mu$ :

$$\mu \geq \sqrt{\frac{n-m}{m(n-1)}}$$

□.

(!!) When  $n \gg m$ , the lower bound is approximately  $\frac{1}{\sqrt{m}}$ .

Thus, bounds <sup>ens</sup> in terms of  $\mu$  work well when  $s \ll \sqrt{m}$ .

(!!) The inequality in the theorem becomes equality when

$$\langle C, \bar{I} \rangle_F = \|C\|_F \sqrt{m}$$

(equality in Cauchy-Schwarz).

This is achieved when  $C$  and  $\bar{I}$  are linearly dependent



i.e.  $\exists \lambda \neq 0 \quad \lambda \in \mathbb{R}$  such that

$$C = AA^T = \lambda I$$

(the rows of  $A$  are orthogonal and have equal length).

On the other hand, we need

$$\sum_{i \neq j} |\langle a_i, a_j \rangle_2|^2 = n(n-1)\mu$$

Since each term is  $\geq 0$  and

satisfies  $|\langle a_i, a_j \rangle_2| \leq \mu$ , the equality is possible only if

$$\langle a_i, a_j \rangle = \mu.$$



Def A system of normalized vectors  $a_i \in \mathbb{R}^m$ ,  $1 \leq i \leq n$ , is called equiangular if there is a number  $\mu > 0$  such that

$$|\langle a_i, a_j \rangle| = \mu, \quad 1 \leq i \neq j \leq n$$

An equiangular system of vectors  $a_1, \dots, a_n \in \mathbb{R}^m$  is called an equiangular tight frame if the ~~(columns?)~~ rows of the matrix

$$A = (a_1, \dots, a_n) \in \mathbb{R}^{m \times n}$$

are orthogonal with equal norm.

Thus, equiangular tight frames are exactly those families for which



we have the equality in the Welsh bound.

Corollary The coherence  $\mu$  of a matrix  $A \in \mathbb{R}^{m \times n}$  is given by

$$\mu = \sqrt{\frac{n-m}{m(n-1)}}$$

iff the columns of  $A$  form an equiangular tight frame

[Ex]. Assume that the entries of  $A \in \mathbb{R}^{m \times n}$ ,  $n > m$  are independently sampled from the normal distribution with mean zero and variance  $\sigma^2 = \frac{1}{m}$ .

Then, if  $S \leq Cm \frac{1}{\log(\frac{n}{m})}$ ,



the matrix  $A$  satisfies  
the RIP and allows the  
reconstruction of an  $s$ -sparse  
vector  $x \in \mathbb{R}^n$  from its observation  
 $b := Ax$  with probability  
 $1 - O(e^{-\gamma n})$   
for some  $\gamma > 0$ .